

LIMIT THEOREMS FOR QUADRATIC FORMS OF MARKOV CHAINS

YVES F. ATCHADÉ AND MATIAS D. CATTANEO

(April 2011)

ABSTRACT. We develop a martingale approximation approach to studying the limiting behavior of quadratic forms of Markov chains. We use the technique to examine the classical and “fixed-b” asymptotic behavior of lag-window estimators for long-run variance estimation, and apply this result to Markov Chain Monte Carlo.

1. INTRODUCTION

This paper deals with quadratic forms given by

$$U_n(h_n) = \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_n(\ell, j) h_n(X_{\ell}, X_j), \quad n \geq 1, \quad (1)$$

for a stochastic process $\{X_n, n \geq 0\}$, weight matrices $w_n : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ and symmetric kernels $h_n : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$. Quadratic forms of possibly time-dependent random variables naturally arise in a variety of statistical and econometric problems, and their large sample properties are of particular importance to develop asymptotically valid inference procedures.

For independent sequences $\{X_n, n \geq 0\}$, the well known Hoeffding decomposition provides a useful approach to studying the asymptotic properties of $U_n(h_n)$ because it decomposes the statistic into two (uncorrelated) martingale sequences, which are then easily handled by standard martingale theory. See, e.g., Serfling (1980) for a review. When the process $\{X_n, n \geq 0\}$ is time-dependent, however, the classical Hoeffding decomposition is not very useful because the resulting representation does not have the desirable martingale property in general. As a consequence, the large sample properties of quadratic forms of time-dependent random variables are typically established in a less systematic way. The most well understood case is the case of a standard U-statistics where h_n does

2000 *Mathematics Subject Classification.* 60J10, 62M10.

Key words and phrases. Central limit theorems, Markov Chains, Markov Chain Monte Carlo, Martingale approximations, Quadratic forms, U-statistics.

Y. F. Atchadé: University of Michigan, Department of Statistics, Ann Arbor, 48109, MI, United States.
E-mail address: yvesa@umich.edu.

Matias D. Cattaneo: University of Michigan, Department of Economics, Ann Arbor, 48109, MI, United States.
E-mail address: cattaneo@umich.edu.

not depend on n and $w_n(\ell, j) = 1$ if $\ell \neq j$ and 0 otherwise (Yoshihara (1976); Eagleson (1979); Dehling and Wendler (2010)). There has been some recent progress. Hsing and Wu (2004) considers $U_n(h)$ where neither h_n nor w_n depends on n , whereas Wu and Shao (2007) studies $U_n(h_n)$ when $h_n(x, y) = h(x, y) = xy$ for a martingale-difference sequence (see also Bhansali et al. (2007) for i.i.d. sequences).

We develop a martingale approximation for $U_n(h_n)$ when $\{X_n, n \geq 0\}$ is a Markov chain. More precisely, we show in Lemma 2.2 under some weak dependence assumption that

$$U_n(h_n) = \theta_n \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_n(\ell, j) + \sum_{\ell=1}^n w_{n,1}(\ell) L_{n,\ell} + \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_n(\ell, j) Q_{n,\ell,j} + \zeta_n,$$

where $\sum_{\ell=1}^n w_{n,1}(\ell) L_{n,\ell}$, $\sum_{\ell=2}^n \sum_{j=1}^{\ell-1} w_n(\ell, j) Q_{n,\ell,j}$ are martingale sequences, $\{\zeta_n, n \geq 1\}$ is a remainder process, and with $\theta_n = \int \int h_n(x, y) \mu(dx) \mu(dy)$, $w_{n,1}(\ell) = \sum_{j=1}^{\ell} w_n(\ell, j) + \sum_{j=\ell}^n w_n(j, \ell)$. Assuming that the kernels h_n are nondegenerate, we use this result to derive some general conditions under which the leading stochastic term in $U_n(h_n)$ is the linear martingale $\sum_{\ell=1}^n w_{n,1}(\ell) L_{n,\ell}$ and we prove that

$$\sigma_n^{-1} \left(U_n(h_n) - \theta_n \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_n(\ell, j) \right) \xrightarrow{w} \mathcal{N}(0, 1),$$

where σ_n^2 is given in (??) and $\xrightarrow{w}$ denotes weak convergence.

As another application we exploit the martingale approximation to study the asymptotic behavior of lag-window estimators of asymptotic variance (see, e.g., Priestley (1981) for detailed discussion and early references). We investigate the classical and “fixed-b” asymptotics of lag-window long-run variance estimators of possibly non-stationary Markov chains. Our results have important implications for Markov Chain Monte Carlo (MCMC) simulations, offering in particular new robust procedures for constructing confidence intervals. Our results also substantially improve on the existing literature on the convergence of lag-windows estimators for Markov chains (Flegal and Jones (2010), Atchade (2010)). The findings from a small simulation study show that the robust confidence intervals derived under the “fixed-b asymptotic” framework have better empirical coverage properties than their counterparts derived under classical approximations in samples of moderate size.

Using Lemma 2.2, we derive a decomposition of the lag-window estimator $\Gamma_{n,b}^2(h)$ that sheds some light on the asymptotic and finite-sample behavior of the estimator, in both cases $c_n = n$ and $c_n = o(n)$. Then we study the asymptotics of $\Gamma_{n,b}^2(h)$ under weaker conditions for nonstationary Markov chain and under conditions that are more easily verifiable. The conditions given below substantially improve on Flegal and Jones (2010) and Atchade (2010). We impose the following assumption.

The paper is organized as follows. The rest of the introduction outlines the general setup and introduces the main notation employed throughout, while Section 2 derives the main martingale approximation method. Section 3 derives the large sample properties of lag-window long-run variance estimators under quite general conditions and, in particular, applies these results to MCMC simulation. All the proofs are presented in Section 4.

1.1. Setup and Notation. We will rely on the following set of notation. Suppose that $(\mathbb{T}, \mathcal{A})$ be an arbitrary measure space. If $W : \mathbb{T} \rightarrow [1, +\infty)$ is a function, the W -norm of a function $f : \mathbb{T} \rightarrow \mathbb{R}$ is defined as $\|f\|_W := \sup_{x \in \mathbb{T}} |f(x)|/W(x)$. The set of measurable functions $f : \mathbb{T} \rightarrow \mathbb{R}$ with finite W -norm is denoted by $\mathcal{L}_W(\mathbb{T})$ or simply $\mathcal{L}_W$ when there is no ambiguity on the space $\mathbb{T}$. For a finite real-valued signed measure ν on $\mathbb{T}$, we denote the W -norm of ν as

$$\|\nu\|_W := \int W(x) |\nu|(dx) = \sup_{|f|_W \leq 1} \left| \int f(x) \nu(dx) \right|,$$

where $|\nu|$ is the total variation measure of ν . We denote $\mathcal{M}_W(\mathbb{T})$ the space of all finite real-valued signed measures ν on $\mathbb{T}$ such that $\|\nu\|_W < \infty$. It is well-known that $(\mathcal{M}_W(\mathbb{T}), \|\cdot\|_W)$ is a Banach space. When the measure space $\mathbb{T}$ is understood, we simply write $\mathcal{M}_W$. We will use the notation $\nu(f)$ to denote the integral $\int f(x) \nu(dx)$.

If μ, ν are two finite signed measures on $(\mathbb{T}, \mathcal{A})$, we denote their product by $\mu\nu$ or $\mu \otimes \nu$. And the product of a finite number k of finite signed measures $\nu_1, \dots, \nu_k$ is denoted by $\otimes_{j=1}^k \nu_j$.

If Q is a transition kernel on $(\mathbb{T}, \mathcal{A})$, and $h : \mathbb{T} \times \mathbb{T} \rightarrow \mathbb{R}$ a bivariate function then Qh is the bivariate function defined by the rule $Qh(x, y) = \int Q(x, dz) h(z, y)$ and Q^2h is defined as $Q^2h(x_1, x_2) = \int Q(x_1, dz_1) \int Q(x_2, dz_2) h(z_1, z_2)$. If $h : \mathbb{T} \rightarrow \mathbb{R}$ is univariate, Qh is defined similarly as $Qh(x) = \int Q(x, dz) h(z)$. Fix Q a Markov kernel, and $V : \mathbb{T} \times \mathbb{T} \rightarrow [1, \infty)$. For $p \geq 1$ and a function $h : \mathbb{T} \times \mathbb{T} \rightarrow \mathbb{R}$, we define

$$\|h\|_{p,V} := \sup_{x,y \in \mathbb{T}} \frac{(\int Q(x, dz) |h(z, y)|^p)^{1/p}}{V(x, y)}.$$

For a univariate function $V : \mathbb{T} \rightarrow [1, \infty)$ and for $h : \mathcal{X} \rightarrow \mathbb{R}$, we define $\|h\|_{p,V}$ similarly as

$$\|h\|_{p,V} := \sup_{x \in \mathbb{T}} V(x)^{-1} \left(\int Q(x, dz) |h(z)|^p \right)^{1/p}.$$

When we use the notation $\|h\|_{p,V}$ below, it will always be with respect to P , the Markov kernel of the reference process $\{X_n, n \geq 0\}$, unless stated otherwise. The following short-range dependence concept will play an important rule.

Definition 1.1. Fix $r \in \mathbb{N}$. For measurable functions $\bar{V}_r \leq \bar{W}_r : \mathbb{T}^r \rightarrow [1, \infty)$, we say that the transition kernel Q with invariant distribution μ satisfies the condition $\mathcal{C}(r, \bar{V}_r, \bar{W}_r)$ if

there exists a finite constant c such that

$$\sum_{j_1 \geq 0} \cdots \sum_{j_r \geq 0} \left\| \bigotimes_{j=1}^r \left(Q^{\ell_j}(x_j, \cdot) - \mu \right) \right\|_{\bar{V}_r} \leq c \bar{W}_r(x_{1:r}), \quad x_{1:r} \in \mathcal{X}^r. \quad (2)$$

Throughout the paper, we denote by c a finite constant which depends solely on the kernel P but whose actual value can change from one equation to the next. In particular c does not depend on the family of function $\{h_n, n \geq 1\}$ considered. Finally, all limits are taken as $n \rightarrow \infty$ unless explicitly noted otherwise.

2. A MARTINGALE APPROXIMATION FOR QUADRATIC FORMS

For notational convenience, we shall write $\bar{\mu}$ to denote the product probability measure $\bar{\mu}(du, dv) = \mu(du)\mu(dv)$, where μ is the invariant distribution of the Markov kernel P . Consider the following assumption.

Assumption A1 For $r = 1, 2$, there exist symmetric measurable functions $\bar{V}_r \leq \bar{W}_r : \mathcal{X}^r \rightarrow [1, \infty)$ such that P satisfies $C(r, \bar{V}_r, \bar{W}_r)$. Furthermore, $P^s \bar{W}_r(x) < \infty$ for all $x \in \mathcal{X}^r$ and all $s \in \{1, \dots, r\}$.

Remark 1. It is always possible to deduce A1 from a univariate short-range dependence assumption. Indeed, if P satisfies $C(1, V_1, W_1)$ and $C(1, V_2, W_2)$, and $PW_1 < \infty$, $PW_2 < \infty$, define $\bar{V}_2(x, y) = V_1(x)V_2(y)$ and $\bar{W}_2(x, y) = W_1(x)W_2(y)$. Then

$$\| (P^n(x, \cdot) - \mu) \bigotimes (P^m(y, \cdot) - \mu) \|_{\bar{V}_2} = \| P^n(x, \cdot) - \mu \|_{V_1} \| P^m(y, \cdot) - \mu \|_{V_2},$$

and therefore A1 holds. But the choice $\bar{V}_2(x, y) = V_1(x)V_2(y)$, $\bar{W}_2(x, y) = W_1(x)W_2(y)$ is just one possibility among many others.

Remark 2. The univariate condition $C(1, V, W)$ holds for geometrically ergodic Markov kernels (that is, kernels P for which $\|P^n(x, \cdot) - \mu\|_V$ converges to zero exponentially fast for some $V \geq 1$). It also holds for sub-geometrically ergodic Markov kernels ($\|P^n(x, \cdot) - \mu\|_V$ converges to zero sub-geometrically) for which the rate of convergence is summable. It is sometimes possible to check the condition $C(1, V, W)$ using Lyapunov drift conditions and their extensions and this has been done for several time series Markov models (Douc et al. (2004); Meitz and Saikkonen (2008); Meyn and Tweedie (2009)).

We show that whenever A1 holds, there exists a martingale approximation to $U_n(h_n)$ that offers a simple route to study the asymptotics of $U_n(h_n)$. For $r \in \{1, 2\}$, the space $\mathcal{M}_{\bar{V}_r}(\mathcal{X}^r)$ of all finite signed measure on $\mathcal{X}^r$ with finite $\|\cdot\|_{\bar{V}_r}$ norm, equipped with the

norm $\|\cdot\|_{\bar{V}_r}$ is a Banach space. Under A1 and for any $x, y \in \mathcal{X}$,

$$\begin{aligned} \bar{R}_1(x; du) &:= \sum_{n \geq 0} (P^n(x, du) - \mu(du)), \\ \text{and } \bar{R}_2(x, y; (du, dv)) &:= \sum_{n_1 \geq 0} \sum_{n_2 \geq 0} (P^{n_1}(x, du) - \mu(du)) \otimes (P^{n_2}(y, dv) - \mu(dv)) \end{aligned}$$

are finite signed measures that belongs respectively to $\mathcal{M}_{\bar{V}_1}(\mathcal{X})$ and $\mathcal{M}_{\bar{V}_2}(\mathcal{X}^2)$. Furthermore we have for all $x, y \in \mathcal{X}$,

$$\|\bar{R}_1(x; \cdot)\|_{\bar{V}_1} \leq c\bar{W}_1(x), \quad \text{and} \quad \|\bar{R}_2(x, y; \cdot)\|_{\bar{V}_2} \leq c\bar{W}_2(x, y). \quad (3)$$

Let $h : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a symmetric measurable function such that $\bar{\mu}(|h|) < \infty$. Denote $\theta = \int \int h(x, y) \mu(dx) \mu(dy)$ and define

$$\bar{h}_1(x) := \int h(x, z) \mu(dz) - \theta, \quad \bar{h}_2(x, y) = h(x, y) - \bar{h}_1(x) - \bar{h}_1(y) - \theta,$$

$$\bar{G}_1(x) := \int \bar{R}_1(x; dz) \bar{h}_1(z), \quad \text{and} \quad \bar{G}_2(x, y) := \int \int \bar{R}_2(x, y; dz_1, dz_2) \bar{h}_2(z_1, z_2), \quad x, y \in \mathcal{X}.$$

We say that h is degenerate when $\bar{h}_1$ is identically zero.

Lemma 2.1. *Assume A1.*

- (1) *Suppose that $\bar{h}_1 \in \mathcal{L}_{\bar{V}_1}$. Then $\bar{G}_1$ is well-defined, $\bar{G}_1 \in \mathcal{L}_{\bar{W}_1}$, and $|\bar{G}_1|_{\bar{W}_1} \leq c|\bar{h}_1|_{\bar{V}_1}$. Furthermore for all $x \in \mathcal{X}$,*

$$\bar{h}_1(x) = \int (\delta_x(dz) - P(x, dz)) \bar{G}_1(z) = \bar{G}_1(x) - P\bar{G}_1(x). \quad (4)$$

If in addition $P\bar{h}_1 \in \mathcal{L}_{\bar{V}_1}$, then $|P\bar{G}_1|_{\bar{W}_1} \leq c|P\bar{h}_1|_{\bar{V}_1}$.

- (2) *Suppose that $\bar{h}_2 \in \mathcal{L}_{\bar{V}_2}$. Then $\bar{G}_2$ is well-defined, $\bar{G}_2 \in \mathcal{L}_{\bar{W}_2}$, and $|\bar{G}_2|_{\bar{W}_2} \leq c|\bar{h}_2|_{\bar{V}_2}$ and for all $x, y \in \mathcal{X}$,*

$$\bar{h}_2(x, y) = \int (\delta_x(dz_1) - P(x, dz_1)) \int (\delta_y(dz_2) - P(y, dz_2)) \bar{G}_2(z_1, z_2). \quad (5)$$

If in addition $P^s \bar{h}_2 \in \mathcal{L}_{\bar{V}_2}$, then $|P^s \bar{G}_2|_{\bar{W}_2} \leq c|P^s \bar{h}_2|_{\bar{V}_2}$ for $s \in \{1, 2\}$.

Proof. See Section 5.1. □

We introduce the functions

$$\Lambda_1(x, y) = \bar{G}_1(y) - P\bar{G}_1(x), \quad x, y \in \mathcal{X}, \quad \text{and}$$

$$\begin{aligned} \Lambda_2(x_1, x_2; y_1, y_2) &= \int (\delta_{y_1}(dz_1) - P(x_1, dz_1)) \int (\delta_{y_2}(dz_2) - P(x_2, dz_2)) \bar{G}_2(z_1, z_2) \\ &= \bar{G}_2(y_1, y_2) - P\bar{G}_2(x_2, y_1) - P\bar{G}_2(x_1, y_2) + P^2\bar{G}_2(x_1, x_2), \quad x_1, x_2, y_1, y_2 \in \mathcal{X}. \end{aligned}$$

Then (4) translates into $\bar{h}_1(x) = \Lambda_1(x, x)$ and (5) can be written as $\bar{h}_2(x, y) = \Lambda_2(x, y, x, y)$. A specially important property of Λ_1, Λ_2 that we rely on in the sequel is the following. For any $x, y, u, v \in \mathcal{X}$,

$$\int P(x, dy) \Lambda_1(x, y) = 0, \quad \text{and} \quad \int P(x, dy) \Lambda_2(u, x, v, y) = \int P(u, dv) \Lambda_2(u, x, v, y) = 0. \quad (6)$$

Remark 3. Equation (5) gives a bivariate Poisson's equation which extends the well known univariate Poisson's equation (4).

Now suppose that we have $\{h_n : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}\}$, a family of symmetric measurable functions such that $\bar{\mu}(|h_n|) < \infty$. We write $\theta_n, \bar{h}_{n,1}, \bar{h}_{n,2}, \bar{G}_{n,1}, \bar{G}_{n,2}, \Lambda_{n,1}$ and $\Lambda_{n,2}$ to denote respectively the quantities $\theta, \bar{h}_1, \bar{h}_2, \bar{G}_1, \bar{G}_2, \Lambda_1$ and Λ_2 defined above with $h = h_n$.

For $1 \leq j \leq \ell \leq n$, we introduce the random variables

$$L_{n,\ell} := \Lambda_{n,1}(X_{\ell-1}, X_\ell), \quad \text{and} \quad Q_{n,\ell,j} := \Lambda_{n,2}(X_{j-1}, X_{\ell-1}, X_j, X_\ell).$$

By the Markov property and (6), we have $\mathbb{E}(L_{n,\ell} | \mathcal{F}_{\ell-1}) = \int P(X_{\ell-1}, dz) \Lambda_{n,1}(z, X_{\ell-1}) = 0$ almost surely, showing that $\{(L_{n,\ell}, \mathcal{F}_\ell), 1 \leq \ell \leq n\}$ is a martingale-difference array. Similarly, assuming $j < \ell$,

$$\mathbb{E}(Q_{n,\ell,j} | \mathcal{F}_{\ell-1}) = \int P(X_{\ell-1}, dz) \Lambda_{n,2}(X_{j-1}, X_{\ell-1}, X_j, z) = 0,$$

almost surely, using (6). This shows that $\{(\sum_{j=1}^{\ell-1} Q_{n,\ell,j}, \mathcal{F}_\ell), 2 \leq \ell \leq n\}$ is a martingale-difference array.

We need the following sequences

$$\begin{aligned} w_{n,1}(\ell) &:= \left\{ \sum_{j=1}^{\ell} w_n(\ell, j) + \sum_{j=\ell}^n w_n(j, \ell) \right\}, \quad \varpi_{n,1}(\ell) := w_n(\ell, j) - w_n(\ell-1, j) \\ \varpi_{n,2}(\ell, j) &:= w_n(\ell, j) - w_n(\ell, j-1), \\ \text{and } \varpi_{n,3}(\ell, j) &:= w_n(\ell, j) + w_n(\ell-1, j-1) - w_n(\ell, j-1) - w_n(\ell-1, j). \end{aligned}$$

Lemma 2.2. *Assume A1 and suppose that $\bar{h}_{n,1} \in \mathcal{L}_{\bar{V}_1}$, and $\bar{h}_{n,2} \in \mathcal{L}_{\bar{V}_2}$. Then*

$$U_n(h_n) = U_{n,0} + \sum_{\ell=1}^n w_{n,1}(\ell) L_{n,\ell} + \sum_{\ell=1}^n w_n(\ell, \ell) Q_{n,\ell,\ell} + \sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_n(\ell, j) Q_{n,\ell,j} + \zeta_n^{(1)} + \zeta_n^{(2)}, \quad (7)$$

where $U_{n,0} = \theta_n \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_n(\ell, j)$, and

$$\zeta_n^{(1)} = w_{n,1}(0) P \bar{G}_{n,1}(X_0) - w_{n,1}(n) P \bar{G}_{n,1}(X_n) + \sum_{\ell=1}^n ((w_{n,1}(\ell) - w_{n,1}(\ell-1)) P \bar{G}_{n,1}(X_{\ell-1}),$$

and

$$\begin{aligned}\zeta_n^{(2)} &= \sum_{\ell=1}^n \sum_{j=1}^{\ell} \varpi_n^{(1)}(\ell, j) (P\bar{G}_{n,2}(X_{\ell-1}, X_j) - P^2\bar{G}_{n,2}(X_{\ell-1}, X_{j-1})) \\ &\quad + \sum_{\ell=1}^n \sum_{j=1}^{\ell} \varpi_n^{(2)}(\ell, j) (P\bar{G}_{n,2}(X_{j-1}, X_{\ell}) - P^2\bar{G}_{n,2}(X_{\ell-1}, X_{j-1})) \\ &\quad + \sum_{\ell=1}^n \sum_{j=1}^{\ell} \varpi_n^{(3)}(\ell, j) P^2\bar{G}_{n,2}(X_{\ell-1}, X_{j-1}) + \epsilon_n,\end{aligned}$$

where

$$\begin{aligned}\epsilon_n &= \sum_{\ell=1}^n \{w_n(\ell, 0)P\bar{G}_{n,2}(X_0, X_{\ell}) - w_n(\ell-1, 0)P^2\bar{G}_{n,2}(X_0, X_{\ell-1})\} \\ &\quad + \sum_{j=1}^n w_n(n, j) (P^2\bar{G}_{n,2}(X_n, X_j) - P\bar{G}_{n,2}(X_n, X_j)) \\ &\quad + \sum_{\ell=1}^n (w_n(\ell-1, \ell)P\bar{G}_{n,2}(X_{\ell-1}, X_{\ell}) - w_n(\ell, \ell)P\bar{G}_{n,2}(X_{\ell}, X_{\ell})).\end{aligned}$$

Proof. See Section 5.2. □

Remark 4. The usefulness of this decomposition comes from the fact that the remainders $\zeta_n^{(1)}$ and $\zeta_n^{(2)}$ involve either single summations or difference sequences of the weights w_n . As a result, these remainders are typically negligible compared to the other terms in the decomposition and one can easily study the asymptotic behavior of $U_n(h_n)$ by focusing on the martingale terms $\sum_{\ell=1}^n w_{n,1}(\ell)L_{n,\ell}$ and $\sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_n(\ell, j)Q_{n,\ell,j}$ and the diagonal term $\sum_{\ell=1}^n w_n(\ell, \ell)Q_{n,\ell,\ell}$.

Remark 5. An interesting question is whether A1 is necessary for (7). Clearly, some sort of short range dependence is needed to obtain a useful martingale approximation. For linear partial sums of a Markov chain, Maxwell and Woodroffe (2000) constructed a martingale approximation using an approximate Poisson's equation solution based on a weighted resolvent kernel. In principle, their idea, which consists in replacing the usual summation by an Abel summation, can be lifted to the bivariate setting. For $\epsilon > 0$, consider the sum

$$\bar{R}_{\epsilon}(x, y; (du, dv)) = \sum_{n \geq 0} \sum_{m \geq 0} \left(\frac{1}{1 + \epsilon} \right)^{n+m+2} (P^n(x, du) - \mu(du)) (P^m(y, dv) - \mu(dv)).$$

Because of the geometric weight $(1 + \epsilon)^{-n-m-2}$, for $\epsilon > 0$, $\bar{R}_{\epsilon}(x, y; \cdot)$ is well defined under weaker conditions than A1. Define $\bar{G}_2^{(\epsilon)}h(x, y) := \int \bar{R}_{\epsilon}(x, y; (du, dv))\bar{h}_2(u, v)$. Then, formally we have

$$(1 + \epsilon)^2 \bar{h}_2(x, y) = \int (\delta_x(dz_1) - P(x, dz_1)) \int (\delta_y(dz_2) - P(y, dz_2)) \bar{G}_2^{(\epsilon)}(z_1, z_2), \quad x, y \in \mathcal{X}. \quad (8)$$

Equation (8) is an approximate bivariate Poisson's equation for h (and P) and can be used in place of the Poisson's equation (5) to derive a martingale approximation for quadratic forms under weaker conditions. We do not pursue this idea further, but we mention this as a promising avenue for future work.

3. APPLICATION: ASYMPTOTIC VARIANCE ESTIMATION

In this section, we use the martingale approximation of Lemma 2.2 to study the asymptotics of lag-windows estimators of asymptotic variance in time series and we apply the results to Markov chain Monte Carlo. Let $h : \mathcal{X} \rightarrow \mathbb{R}$ be a measurable function such that $\mu(|h|^2) < \infty$. We assume without any loss of generality that $\mu(h) = 0$. We are interested in the estimation of the long-run variance (or the asymptotic variance) of h defined as:

$$\sigma^2(h) = \text{Var}_\mu(h(X_0)) + 2 \sum_{\ell \geq 1} \text{Cov}_\mu(h(X_0), h(X_\ell)), \quad (9)$$

which plays a role in time series analysis and in Markov Chain Monte Carlo. A classical estimator for $\sigma^2(h)$ is the lag-windows estimators defined as

$$\Gamma_{n,b}^2(h) := \gamma_{n,0} + 2 \sum_{k=1}^{n-1} w_b(kc_n^{-1}) \gamma_{n,k}, \quad (10)$$

where $\gamma_{n,k} := n^{-1} \sum_{j=1}^{n-k} (h(X_j) - \mu_n(h)) (h(X_{j+k}) - \mu_n(h))$ is the k -th order sample autocovariance with $\mu_n(h) = n^{-1} \sum_{j=1}^n h(X_j)$, w_b is a weight function (with a parameter b) and $\{c_n, n \geq 1\}$ is an increasing sequence of positive numbers. We refer the reader to Priestley (1981) for detailed discussion and early references on lag-windows estimators. We consider weight functions with the following properties.

Assumption W: For $b > 0$, $w_b : \mathbb{R} \rightarrow [0, 1]$ is a continuous function with support $[0, b]$, of class $\mathcal{C}^2$ on the interval $(0, b)$, such that $w_b(b) = 0$ and $w_b(0) = 1$.

This assumption allows for the use of all commonly employed weighting functions, including the Bertlett and Parzen kernels. When $w_b(x) = w(x/b)$, an equivalent parametrization of $\Gamma_{n,b}^2(h)$ is $\Gamma_{n,b}^2(h) = \gamma_{n,0} + 2 \sum_{k=1}^{n-1} w(kc_n^{-1}) \gamma_{n,k}$, with $c_n \leftarrow bc_n$.

We shall now use Lemma 2.2 to derive a decomposition of $\Gamma_{n,b}(h)$ that sheds some new light on the finite sample and asymptotic properties of the estimator. We impose the following ergodicity assumption.

Assumption A2 There exist measurable functions $V_k : \mathcal{X} \rightarrow [1, \infty)$ ($k = 1, 2, 3$), $V_1 \leq V_2, V_2^2 \leq V_3$, such that $PV_3(x) < \infty$ for all $x \in \mathcal{X}$, and P satisfies the assumptions $\mathcal{C}(1, V_1, V_2)$ and $\mathcal{C}(1, V_2^2, V_3)$. Furthermore there exists $q > 1$ such that

$$\sup_{n \geq 0} \mathbb{E}(V_3^q(X_n)) < \infty. \quad (11)$$

A2 implies A1 with $\bar{V}_1 = V1$, $\bar{W}_1 = V2$, $\bar{V}_2(x, y) = V_1(x)V_1(y)$ and $\bar{W}_2(x, y) = V_2(x)V_2(y)$. Define the partial sums $S_{n,k} := \sum_{j=k+1}^{k+n} h(X_j)$, and the weight $w_{n,b}(0) = n^{-1}$ and $w_{n,b}(k) = 2n^{-1}w_b(kc_n^{-1})$ for $k > 0$.

We can rewrite $\gamma_{n,k}$ as

$$\gamma_{n,k} = n^{-1} \sum_{j=1}^{n-k} h(X_j)h(X_{j+k}) + n^{-3}(n-k)S_{n,0}^2 - n^{-2}S_{n,0}(S_{n-k,0} + S_{n-k,k}),$$

so that

$$\Gamma_{n,b}^2(h) = \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_{n,b}(\ell-j)h(X_j)h(X_{\ell}) + R_n, \quad (12)$$

where

$$\begin{aligned} R_n = & 2n^{-2}S_{n,0}^2 \sum_{k=1}^{n-1} w_b(kc_n^{-1}) \left(1 - \frac{k}{n}\right) \\ & - 2n^{-2}S_{n,0} \left(\sum_{j=2}^n h(X_j) \sum_{k=1}^{j-1} w_b(kc_n^{-1}) + \sum_{j=1}^{n-1} h(X_j) \sum_{k=1}^{n-j} w_b(kc_n^{-1}) \right) - n^{-2}S_{n,0}^2. \end{aligned} \quad (13)$$

If we set aside the term R_n , the expression (12) is of the form (1) with $h_n(x, y) = h(x)h(y)$ and $w_n(\ell, j) = w_{n,b}(\ell-j)$. Here we have $h_{n,1}(x) = \int h(x)h(y)\mu(dy) = 0$, $\theta_n = 0$, and $h_{n,2}(x, y) = h(x)h(y)$. Therefore $L_{n,\ell} = 0$. Define

$$G(x) := \sum_{j \geq 0} P^j h(x), \quad \text{and} \quad PG(x) = \int P(x, dz)G(z), \quad x \in \mathcal{X}.$$

Then $\bar{G}_2(x, y) = G(x)G(y)$, $P\bar{G}_2(x, y) = PG(x)G(y)$, and $P^2\bar{G}_2(x, y) = PG(x)PG(y)$. Therefore

$$Q_{n,\ell,j} = Q_{\ell}Q_j, \quad \text{where} \quad Q_{\ell} = G(X_{\ell}) - PG(X_{\ell-1}).$$

As above, $\{(Q_{\ell}, \mathcal{F}_{\ell}), \ell \geq 1\}$ is a martingale: $\mathbb{E}(Q_{\ell}|\mathcal{F}_{\ell-1}) = 0$. From Lemma 2.2 we obtain the following.

Theorem 3.1. *Assume (A2) and (W) and $h \in \mathcal{L}_{V_1}$. For all $n \geq 1$,*

$$\Gamma_{n,b}^2(h) = n^{-1} \sum_{\ell=1}^n Q_{\ell}^2 + \sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_{n,b}(\ell-j)Q_{\ell}Q_j + R_n + \zeta_n. \quad (14)$$

Furthermore, there exist $p > 1$ and a finite constant c such that for all $n \geq 3$,

$$\begin{aligned} \mathbb{E}^{1/p}(|\zeta_n|^p) &\leq cc_n^{-1+\frac{1}{2}\vee\frac{1}{p}}, \quad \mathbb{E}^{1/p}(|R_n|^p) \leq cn^{-1}c_n, \\ \text{and} \quad \mathbb{E}^{1/p} \left(\left| \sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_{n,b}(\ell-j)Q_{\ell}Q_j \right|^p \right) &\leq c \left(\frac{c_n}{n} \right)^{\frac{1}{2}} n^{-\frac{1}{2}+\frac{1}{p}\vee\frac{1}{2}}. \end{aligned}$$

Proof. See Section 5.3. □

A clearer picture of the behavior of the lag-window estimator emerges from Theorem 3.1. For $p \geq 2$, we have

$$\Gamma_{n,b}^2(h) = \underbrace{n^{-1} \sum_{\ell=1}^n Q_\ell^2}_{O_p(1)} + \underbrace{\sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_{n,b}(\ell-j) Q_\ell Q_j}_{O_p(\sqrt{\frac{c_n}{n} + \frac{c_n}{n}})} + R_n + \underbrace{\zeta_n}_{o_p(1)}, \quad (15)$$

By the law of large numbers for Markov chain the term $n^{-1} \sum_{\ell=1}^n Q_\ell^2$ converges to $\sigma^2(h)$. As the result, Theorem 3.1 implies that $\Gamma_{n,b}^2(h)$ converges in probability to $\sigma^2(h)$ provided $c_n = o(n)$ and $p \geq 2$ (for $1 < p < 2$, specific rate assumption on c_n might be needed). However, it is well documented that $\Gamma_{n,b}(h)$ often has poor finite-sample properties in estimating $\sigma^2(h)$, particularly for highly correlated time-series. The decomposition (15) also gives some insight as to why. For $c_n = o(n)$, both terms $R_n + \sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_{n,b}(\ell-j) Q_\ell Q_j$ and ζ_n converge to zero but at antagonistic rates. If $c_n \approx n$, then $\zeta_n \approx O_P(n^{-1/2})$ but then $R_n + \sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_{n,b}(\ell-j) Q_\ell Q_j \approx O(1)$. Whereas for $c_n \ll n$, the convergence of ζ_n is slow ($\zeta_n = O_P(c_n^{-1/2})$) but $R_n + \sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_{n,b}(\ell-j) Q_\ell Q_j$ vanishes quickly.

When the goal is to construct confidence interval for $\mu(h)$ (and we are not interested in estimating $\sigma^2(h)$ per se), it has been suggested to use the lag-window estimator $\Gamma_{n,b}^2(h)$ with $c_n = n$, the so-called “fixed-b asymptotics” (Neave (1970); Kiefer and Vogelsang (2005)). With $c_n = n$, $\Gamma_{n,b}^2(h)$ no longer converges to $\sigma^2(h)$, but as it turns out, asymptotically valid confidence intervals can still be derived for $\mu(h)$. We have the following.

Theorem 3.2. *Under the assumption of Theorem 3.1, the following holds true.*

- (1) *If $p \geq 2$ and $c_n = o(n)$, then $\Gamma_{n,b}^2(h)$ converges in probability to $\sigma^2(h)$. Furthermore, assuming $\Gamma_{n,b}^2(h) > 0$ almost surely,*

$$\{n\Gamma_{n,b}^2(h)\}^{-1/2} \sum_{j=1}^n (h(X_j) - \mu(h)) \xrightarrow{w} \mathcal{N}(0, 1).$$

- (2) *Let $\{B(t), 0 \leq t \leq 1\}$ be the standard Brownian motion. If $c_n = n$, then $\Gamma_{n,b}^2(h) \xrightarrow{w} \sigma^2(h)K_b$, where*

$$K_b = 1 + 2 \int_0^1 \int_0^t w_b(t-s) dB(s) dB(t) - 2B(1) \int_0^1 g_b(t) dB(t) + 2B^2(1) \int_0^1 (1-t) w_b(t) dt,$$

where $g_b(t) = \int_0^t w_b(u) du + \int_0^{1-t} w_b(u) du$. Furthermore, assuming $\Gamma_{n,b}^2(h) > 0$ almost surely,

$$\{n\Gamma_{n,b}^2(h)\}^{-1/2} \sum_{j=1}^n (h(X_j) - \mu(h)) \xrightarrow{w} \frac{B(1)}{\sqrt{K_b}}.$$

Proof. See Section 5.4. □

By Theorem 3.2 (1) an asymptotically valid $(1 - \alpha)$ -confidence interval for $\mu(h)$ is

$$\mu_n(h) \pm z_{1-\alpha/2} \frac{\hat{\sigma}_n(h)}{\sqrt{n}}, \quad (16)$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of the standard normal distribution and where $\hat{\sigma}_n(h) = \sqrt{\Gamma_{n,b}^2(h)}$, with $b = 1$, $c_n = o(n)$. Typical choice of c_n includes $c_n = n^{-\delta}$, $\delta \in (0, 1)$ typically around 0.5. Theorem 3.1 (2) provides another asymptotically valid confidence interval for $\mu(h)$:

$$\mu_n(h) \pm t_{1-\alpha/2} \frac{\tilde{\sigma}_n(h)}{\sqrt{n}}, \quad (17)$$

where $t_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of the distribution of $B(1)/\sqrt{K_b}$ and where $\tilde{\sigma}_n(h) = \sqrt{\Gamma_{n,b}^2(h)}$, with $c_n = bn$, with $b \in (0, 1)$.

Although the limiting distribution $B(1)/\sqrt{K_b}$ is non-standard, it can be simulated, for example by Euler discretization of the stochastic integrals in K_b . We report in Table 1 the 95% quantiles of the distribution of $B(1)/\sqrt{K_b}$ using $w_b(x) = \mathbf{1}_{(0,b)}(x)$, $w_b(x) = (1 - x/b)\mathbf{1}_{(0,b)}(x)$ and $w_b(x) = (1 - (x/b)^2)\mathbf{1}_{(0,b)}(x)$, and for different values of b , based on 10,000 replications of $B(1)/\sqrt{K_b}$. The distribution departs further from the standard normal distribution as b increases.

	$w_b(x) = 1 - x/b$	$w_b(x) = 1 - (x/b)^2$	$w_b(x) = \mathbf{1}_{(0,1)}(x/b)$
$b = 0.3$	2.828	4.134	5.496
$b = 0.5$	3.557	6.580	6.299
$b = 0.9$	4.735	12.575	13.045

TABLE 1. 0.975-quantile of the distribution of $B(1)/\sqrt{K_b}$.

In the next simulation examples, we compare the finite sample properties of these two confidence intervals in terms of coverage probability and interval length. All the simulations are performed using the Bartlett kernel $w(x) = 1 - x$.

3.1. Illustration: the Garch(1,1) model. Consider the linear GARCH(1,1) model defined as follows. $h_0 \in (0, \infty)$, $u_0 \sim \mathcal{N}(0, h_0)$ and for $n \geq 1$

$$\begin{aligned} u_n &= h_n^{1/2} \epsilon_n \\ h_n &= \omega + \beta h_{n-1} + \alpha u_{n-1}^2, \end{aligned}$$

where $\{\epsilon_n, n \geq 0\}$ is i.i.d. $\mathcal{N}(0, 1)$ and $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$. We assume that α, β satisfy

C1 There exists $\nu > 0$ such that

$$\mathbb{E}[(\beta + \alpha Z^2)^\nu] < 1, \quad Z \sim \mathcal{N}(0, 1). \quad (18)$$

It is shown by Meitz and Saikkonen (2008) (Theorem 2) that under (18) the joint process $\{(u_n, h_n), n \geq 0\}$ is a ϕ -irreducible aperiodic Markov chain that admits an invariant distribution and is geometrically ergodic with a drift function $V(u, h) = 1 + h^\nu + |u|^{2\nu}$. Therefore for $\nu \geq 2$, A2 holds with $V_1 = V_2 = V^{1/2}$, and $V_3 = V$. We are interested in a confidence interval for $\mu(h)$ where $h(u) = u^2$ which belongs to $\mathcal{L}_{V_1}$. The exact value is $\mu(h) = \omega(1 - \alpha - \beta)^{-1}$.

For the simulations we set $\omega = 1$, $\alpha = 0.1$, $\beta = 0.7$ which gives $\mu(h) = 5$. We compare the confidence intervals (16) and (17) by computing (by Monte Carlo) their coverage probabilities and average lengths. The comparison is performed using sample paths of length 60,000 from the GARCH(1,1) Markov chain. The results are plotted in Figure 1 and shows across the board better coverage probability of the fixed-b confidence interval but, as expected, at the expense of a slightly wider confidence intervals.

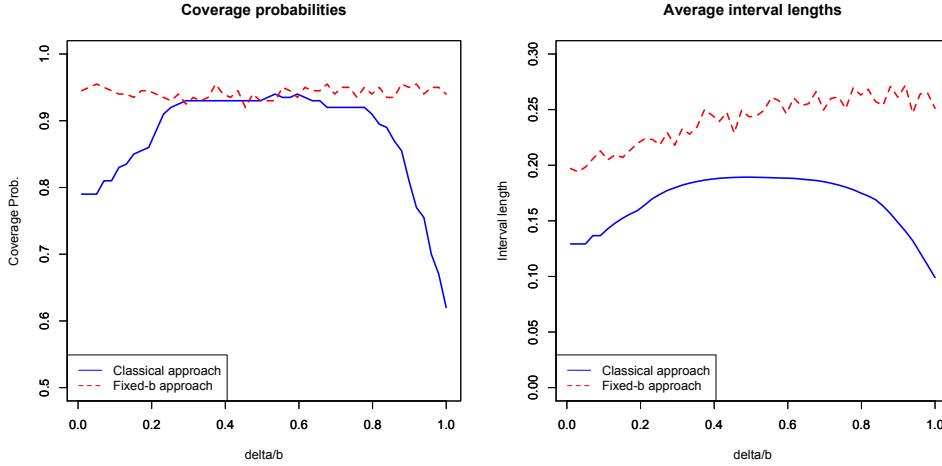


Figure 1: Coverage probabilities plots for various values of δ (classical confidence interval) and b (fixed-b confidence interval).

3.2. Markov Chain Monte Carlo. Markov Chain Monte Carlo (MCMC) is a popular computational tools to obtain random samples from intractable and high-dimensional distributions (see e.g. Roberts and Rosenthal (2004) for a survey and for additional references).

Suppose that we interested in sampling from the probability measure μ and compute the integral $\mu(h) = \int h(x)\mu(dx)$. Let $\{X_n, n \geq 0\}$ be a Markov chains with transition kernel P , invariant distribution μ and initial distribution ρ . By simulating the Markov chain, we approximate $\mu(h)$ by the Monte Carlo average $\mu_n(h) = n^{-1} \sum_{k=1}^n h(X_k)$. Furthermore, under A2, $\lim_{n \rightarrow \infty} n^{1/2} \text{Var}(\mu_n(h)) = \sigma^2(h)$, as given by (9), and a central limit theorem holds: $n^{-1/2} \sum_{k=1}^n (h(X_k) - \pi(h)) \xrightarrow{w} N(0, \sigma^2(h))$. Therefore (16) and (17) provide two valid confidence intervals for $\mu(h)$. We compare the coverage probabilities and average

interval lengths of these two confidence interval procedures with the following simulation example.

3.2.1. Illustration: a Poisson regression model. We undertake the comparison using a log-linear model taken from Gelman et al. (2004). For $e = 1, \dots, N_e$ and $p = 1, \dots, N_p$, the variables y_{ep} are conditionally independent given $(\{\beta_p\}, \{\varepsilon_{ep}\}) \in \mathbb{R}^{N_p} \times \mathbb{R}^{N_e N_p}$, with conditional distribution

$$y_{ep} \sim \mathcal{P}\left(n_{ep} e^{\mu + \alpha_e + \beta_p + \varepsilon_{ep}}\right), \quad e = 1, \dots, N_e, \quad p = 1, \dots, N_p, \quad (19)$$

where $\mathcal{P}(\lambda)$ is the Poisson distribution with parameter λ . In the above display, $\{n_{ep}\}$ is a deterministic baseline covariate, and $\mu \in \mathbb{R}$, $\{\alpha_e\} \in \mathbb{R}^{N_e}$ are parameters. We assume that $\{\beta_p\}$ and $\{\varepsilon_{ep}\}$ are independent with distributions

$$\beta_p \stackrel{iid}{\sim} N(0, \sigma_\beta^2), \quad \varepsilon_{ep} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2), \quad e = 1, \dots, N_e, \quad p = 1, \dots, N_p, \quad (20)$$

for some parameters $\sigma_\beta^2 > 0$, $\sigma_\varepsilon^2 > 0$. We assume a diffuse prior for $(\mu, \alpha, \sigma_\beta^2, \sigma_\varepsilon^2)$ ($\sigma_\varepsilon^2 > 0, \sigma_\beta^2 > 0$) with the additional constraint that $\alpha_{N_e} = -\sum_{k=1}^{N_e-1} \alpha_k$. Let $\theta = (\mu, \alpha, \beta, \epsilon, \sigma_\varepsilon^2, \sigma_\beta^2) \in \mathbb{R}^{3+N_e-1+(N_p+1)N_e}$. The posterior distribution of θ given $\mathcal{D} = (y_{ep}, n_{ep})$ takes the form

$$\pi(\theta|\mathcal{D}) \propto \exp \left\{ \sum_{e,p} y_{e,p} (\mu + \alpha_e + \beta_p + \epsilon_{e,p}) - n_{ep} e^{\mu + \alpha_e + \beta_p + \epsilon_{ep}} - \frac{N_e N_p}{2} \log \sigma_\varepsilon^2 - \frac{N_p}{2} \log \sigma_\beta^2 - \frac{1}{2\sigma_\varepsilon^2} \sum_{e,p} \epsilon_{e,p}^2 - \frac{1}{2\sigma_\beta^2} \sum_{p=1}^{N_p} \beta_p^2 \right\}. \quad (21)$$

This posterior distribution is typical of probability distributions for which MCMC is useful. We set $N_e = 3$ and $N_p = 20$. Suppose that we are interested in a confidence interval for the posterior mean of the parameter α_1 , i.e. $\int \alpha_1 \pi(\theta|\mathcal{D}) d\theta$. To compare the two confidence intervals methods described above, we generate an artificial dataset with $(\alpha_1, \alpha_2, \mu, \sigma_\varepsilon^2, \sigma_\beta^2) = (0.35, 0.15, -1.0, 0.1, 0.3)$. We run a preliminary MCMC sampler for 6 millions (6×10^6) iterations and compute its sample mean. We obtain $\bar{\alpha}_1 = 0.3309$. We take this value to be $\int \alpha_1 \pi(\theta) d\theta$.

To compare the two confidence interval methods, we use a Random Walk Metropolis (RWM) algorithm with proposal kernel $\mathcal{N}(0, \kappa \Sigma)$ where κ and Σ are selected (from a preliminary simulation) to yield a reasonably good mixing of the chain. We run the MCMC sampler for 60,000 iterations and discard the first 10,000 iterations as burn-in. We repeat the simulations 200 times in order to estimate the coverage probabilities and interval lengths. The results are given in Figure 2. We find again that in terms of finite sample behavior, the fixed-b confidence interval is more robust to the choice of b .

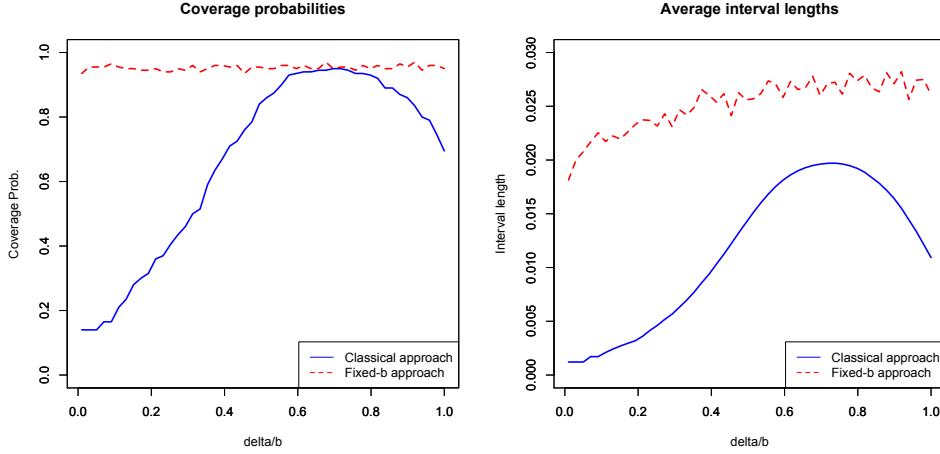


Figure 2: Coverage probability and confidence interval length for α_1 and for different values of b and δ .

4. APPLICATION: A CLT FOR U-STATISTICS WITH VARYING KERNELS

We consider in this section the special case where $w_n(\ell, \ell) = 0$ and $w_n(\ell, j) = 1$ if $\ell \neq j$ resulting in the U-statistic with varying kernels

$$U_n(h_n) = \sum_{\ell=2}^n \sum_{j=1}^{\ell-1} h_n(X_\ell, X_j).$$

We illustrate another application of Lemma 2.2, by deriving a CLT for $U_n(h_n)$. U-statistics and U-statistics with varying kernels play an important role in nonparametric and semi-parametric statistics. In the present case, it is easy to verify that under A1, Lemma 2.2 reduces to

$$U_n(h_n) = \binom{n}{2} \theta_n + (n-1) \sum_{\ell=1}^n L_{n,\ell} + \sum_{\ell=2}^n \sum_{j=1}^{\ell-1} Q_{n,\ell,j} + \zeta_n^{(1)} + \zeta_n^{(2)}.$$

We impose the following assumption

B1 For $p = 2$,

$$\sup_{\ell \geq 0} \mathbb{E}(\bar{W}_1^p(X_\ell)) < \infty, \quad \text{and} \quad \sup_{\ell, j \geq 0} \mathbb{E}(\bar{W}_2^p(X_\ell, X_j)) < \infty. \quad (22)$$

Notice from the Poisson equation (4) that we can also rewrite $L_{n,\ell}$ as $L_{n,\ell} = \bar{h}_{n,1}(X_\ell) + P\bar{G}_1(X_\ell) - P\bar{G}_1(X_{\ell-1})$. And assuming that $P\bar{h}_{n,1} \in \mathcal{L}_{\bar{V}_1}$, we get the bound

$$|L_{n,\ell}| \leq |\bar{h}_{n,1}(X_\ell)| + c|P\bar{h}_{n,1}|_{\bar{V}_1} (\bar{W}_1(X_{\ell-1}) + \bar{W}_1(X_\ell)). \quad (23)$$

We have a similar bound for $Q_{n,\ell,j}$. By definition, $Q_{n,\ell,j} = \bar{h}_{n,2}(X_\ell, X_j) - P\bar{G}_{n,2}(X_{\ell-1}, X_j) - P\bar{G}_{n,2}(X_{j-1}, X_\ell) + P^2\bar{G}_{n,2}(X_{\ell-1}, X_{j-1})$ and assuming that $P\bar{h}_{n,2}, P^2\bar{h}_{n,2} \in \mathcal{L}_{\bar{V}_2}$, one obtains

$$|Q_{n,\ell,j}| \leq |\bar{h}_{n,2}(X_\ell, X_j)| + c|P\bar{h}_{n,2}|_{\bar{V}_2} (\bar{W}_2(X_{\ell-1}, X_j) + \bar{W}_2(X_{j-1}, X_\ell)) + c|P^2\bar{h}_{n,2}|_{\bar{V}_2} \bar{W}_2(X_{\ell-1}, X_{j-1}). \quad (24)$$

We will use (23) and (24) to derive bounds on the p -th moment of $|L_{n,\ell}|$ and $|Q_{n,\ell,j}|$, respectively.

We recall from Lemma 2.1 that $|P\bar{G}_{n,1}(x)| \leq c|P\bar{h}_{n,1}|_{\bar{V}_1} \bar{W}_1(x)$, and for $s \in \{1, 2\}$, $|P^s\bar{G}_{n,2}(x, y)| \leq c|P^s\bar{h}_{n,2}|_{\bar{V}_2} \bar{W}_2(x, y)$. Using this in combination with B1 and the expression of $\zeta_n^{(1)}$ and $\zeta_n^{(2)}$ in Lemma 2.2, it follows that

$$\mathbb{E}^{1/2}(|\zeta_n^{(1)}|^2) \leq cn\kappa_{n,1}, \quad \text{and} \quad \mathbb{E}^{1/2}(|\zeta_n^{(2)}|^2) \leq cn\kappa_{n,2}, \quad (25)$$

where

$$\kappa_{n,1} := |P\bar{h}_{n,1}|_{\bar{V}_1}, \quad \text{and} \quad \kappa_{n,2} := \|\bar{h}_{n,2}\|_{\bar{V}_2} + |P^2\bar{h}_{n,2}|_{\bar{V}_2}.$$

We impose an addition stability assumptions.

B2 With $\bar{W}_2$ as in A1, suppose that $P\bar{W}^2 \leq c\bar{W}^2$ and there exists measurable functions $\bar{U}_1 \leq \bar{V}_1 : \mathcal{X} \rightarrow [1, \infty)$ and a symmetric measurable function $\bar{U}_2 : \mathcal{X} \times \mathcal{X} \rightarrow [1, \infty)$ such that P satisfies C(1, $\bar{U}_1, \bar{V}_1$) and for all $m \geq 0$,

$$\begin{aligned} & \int P(x_{j-1}, dx_j) \int P^m(x_j, dx_{\ell-1}) \\ & \quad \times (\bar{W}(x_{\ell-1}, x_j) + \bar{W}(x_{\ell-1}, x_{j-1})) (\bar{W}(x_{\ell-1}, x_k) + \bar{W}(x_{\ell-1}, x_{k-1})) \\ & \leq c\bar{U}_1(x_{j-1})\bar{U}_2(x_{k-1}, x_k). \end{aligned} \quad (26)$$

Futhermore, we assume that

$$\sup_{\ell \geq 1} \mathbb{E}(\bar{V}_1(X_\ell)\bar{U}(X_\ell, X_{\ell-1})) < \infty.$$

B3 For $\bar{W}_1$ as in A1, suppose that $P\bar{W}_1^2 \leq c\bar{W}_1^2$ for some finite constant c and there exists $\bar{\bar{W}}_1 : \mathcal{X} \rightarrow [1, \infty)$, $\bar{W}_1^2 \leq \bar{\bar{W}}_1$ such that P satisfies C(1, $\bar{W}_1^2, \bar{\bar{W}}_1$) and $\sup_{n \geq 0} \mathbb{E}(\bar{\bar{W}}_1^q(X_n)) < \infty$ for some $q > 1$.

Remark 6. For example if $\bar{W}(x, y) = W(x)W(y)$ and $P^k W \leq cW$ for all $k \geq 0$ and $PW^4 \leq cW^4$, then $P^k \bar{W}^2 \leq c\bar{W}^2$ for all $k \geq 0$, and (26) holds with $\mathcal{U}_1(x) = W^4(x)$ and $\bar{U}(x, y) = \bar{W}(x, y)$.

Define

$$\begin{aligned}\sigma_{n,1}^2 &:= \int \{\mu P\}(dx, dy) L_n^2(x, y) \\ &= \text{Var}_\mu(h_{n,1}(X_0)) + 2 \sum_{\ell \geq 1} \text{Cov}_\mu(h_{n,1}(X_0), h_{n,1}(X_\ell)), \quad \text{and} \quad \sigma_n^2 := n(n-1)^2 \sigma_{n,1}^2.\end{aligned}$$

Theorem 4.1. *Assume A1, B1-B3 and let $\{h_n, n \geq 1\}$ be such that*

$$\kappa_{n,1} + \kappa_{n,2} = o\left(n^{1/2} \sigma_{n,1}\right), \quad \text{and} \quad \|\bar{h}_{n,1}\|_{2q, \bar{V}_1} = o\left(n^{\frac{1}{2}(1-\frac{1}{q})} \sigma_{n,1}\right). \quad (27)$$

Then $\sigma_n^{-1} [U_n(h_n) - \theta_n \binom{n}{2}]$ converges weakly to $\mathcal{N}(0, 1)$.

Proof. See Section 5.5. □

Remark 7. Theorem 4.1 describes the limiting behavior of $U_n(h_n)$ in the case where the kernels h_n are not degenerate and the quadratic term $\sum_{\ell=1}^n \sum_{j=1}^{\ell-1} Q_{n,\ell,j}$ is negligible. Notice that when $h_n = h$ does not depend on n and h is not degenerate, then the quadratic term $\sum_{\ell=1}^n \sum_{j=1}^{\ell-1} Q_{n,\ell,j}$ is necessarily negligible (the condition $\kappa_{n,2} = o(n^{1/2} \sigma_{n,1})$ automatically holds). In general, the quadratic term needs not be negligible. In which case a correct account of the limiting behavior of $U_n(h_n)$ will then require a joint study the two martingales $\sum_{\ell=1}^n \gamma_{n,\ell} L_{n,\ell}$ and $\sum_{\ell=1}^n \sum_{j=1}^{\ell-1} Q_{n,\ell,j}$. We refer the reader to ? where some results have been derived along those lines and using the same techniques as above.

5. PROOFS

5.1. Proof Lemma 2.1. We only prove (2), (1) is similar. That $\bar{G}_2 \in \mathcal{L}_{\bar{W}_2}$, and $|\bar{G}_2|_{\bar{W}_2} \leq c|\bar{h}_2|_{\bar{V}_2}$ follows from the second part of (3). Set

$$\pi_{n,m}(x, y; (du, dv)) = (P^n(x, du - \mu(du)) \otimes (P^m(y, dv) - \mu(dv))).$$

By A1 and dominated convergence, it is easy to show that the rhs of (5) is well-defined and can be written as

$$\begin{aligned}\bar{G}_2(x, y) - P\bar{G}_2(y, x) - P\bar{G}_2(x, y) + P^2\bar{G}_2(x, y) \\ = \lim_{N, M \rightarrow \infty} \sum_{n=0}^N \sum_{m=0}^M \int (\delta_x(dz_1) - P(x, dz_1)) \int (\delta_y(dz_2) - P(y, dz_2)) \pi_{n,m} \bar{h}_2(z_1, z_2), \\ = \lim_{N, M \rightarrow \infty} \{ \bar{h}_2(x, y) - \pi_{N+1,0} \bar{h}_2(x, y) - \pi_{0,M+1} \bar{h}_2(x, y) + \pi_{N+1,M+1} \bar{h}_2(x, y) \} = \bar{h}_2(x, y),\end{aligned}$$

proving (5). The bound $|P^s \bar{G}_2|_{\bar{W}_2} \leq c|P^s \bar{h}_2|_{\bar{V}_1}$ is obtained by showing in a similar way that

$$P^s \bar{G}_2(x, y) = \lim_{N, M \rightarrow \infty} \sum_{n=0}^N \sum_{m=0}^M \pi_{n,m} \{P^s \bar{h}_2\}(x, y).$$

□

5.2. Proof Lemma 2.2. From the definition, we have $h_n(x, y) = \theta_n + \bar{h}_{n,1}(x) + \bar{h}_{n,1}(y) + \bar{h}_{n,2}(x, y)$, and we deduce after some rearrangements that

$$U_n(h_n) = U_{n,0} + \sum_{\ell=1}^n w_{n,1}(\ell) \bar{h}_{n,1}(X_\ell) + \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_n(\ell, j) \bar{h}_{n,2}(X_\ell, X_j).$$

The handling of the term $\sum_{\ell=1}^n w_{n,1}(\ell) \bar{h}_{n,1}(X_\ell)$ is standard. Using (4), we obtain

$$\sum_{\ell=1}^n w_{n,1}(\ell) \bar{h}_{n,1}(X_\ell) = \sum_{\ell=1}^n w_{n,1}(\ell) L_{n,\ell} + \zeta_n^{(1)}, \quad n \geq 1$$

where

$$\zeta_n^{(1)} = w_{n,1}(0) P\bar{G}_{n,1}(X_0) - w_{n,1}(n) P\bar{G}_{n,1}(X_n) + \sum_{\ell=1}^n (w_{n,1}(\ell) - w_{n,1}(\ell-1)) P\bar{G}_{n,1}(X_{\ell-1}).$$

Using (5), we write

$$\begin{aligned} \bar{h}_{n,2}(X_\ell, X_j) &= \Lambda_{n,2}(X_j, X_\ell, X_j, X_\ell) = Q_{n,\ell,j} + \Lambda_{n,2}(X_j, X_\ell, X_j, X_\ell) - \Lambda_{n,2}(X_{j-1}, X_{\ell-1}, X_j, X_\ell) \\ &= Q_{n,\ell,j} + (P\bar{G}_{n,2}(X_{\ell-1}, X_j) - P\bar{G}_{n,2}(X_\ell, X_j)) \\ &\quad + (P\bar{G}_{n,2}(X_{j-1}, X_\ell) - P\bar{G}_{n,2}(X_j, X_\ell)) + (P^2\bar{G}_{n,2}(X_j, X_\ell) - P^2\bar{G}_{n,2}(X_{j-1}, X_{\ell-1})). \end{aligned}$$

Rearranging the terms, it is easy to verify that

$$\begin{aligned} \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_n(\ell, j) \bar{h}_{n,2}(X_\ell, X_j) &= \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_n(\ell, j) Q_{n,\ell,j} \\ &\quad + \sum_{\ell=1}^n \sum_{j=1}^{\ell} \varpi_n^{(1)}(\ell, j) (P\bar{G}_{n,2}(X_{\ell-1}, X_j) - P^2\bar{G}_{n,2}(X_{j-1}, X_{\ell-1})) \\ &\quad + \sum_{\ell=1}^n \sum_{j=1}^{\ell} \varpi_n^{(2)}(\ell, j) (P\bar{G}_{n,2}(X_{j-1}, X_\ell) - P^2\bar{G}_{n,2}(X_{j-1}, X_{\ell-1})) \\ &\quad + \sum_{\ell=1}^n \sum_{j=1}^{\ell} \varpi_n^{(3)}(\ell, j) P^2\bar{G}_{n,2}(X_{j-1}, X_{\ell-1}) + \epsilon_n, \quad (28) \end{aligned}$$

where ϵ_n is comprised of the remainder telescoping sums. We obtain

$$\begin{aligned} \epsilon_n &= \sum_{\ell=1}^n \{ w_n(\ell, 0) P\bar{G}_{n,2}(X_0, X_\ell) - w_n(\ell-1, 0) P^2\bar{G}_{n,2}(X_0, X_{\ell-1}) \} \\ &\quad + \sum_{j=1}^n w_n(n, j) (P^2\bar{G}_{n,2}(X_n, X_j) - P\bar{G}_{n,2}(X_n, X_j)) \\ &\quad + \sum_{\ell=1}^n (w_n(\ell-1, \ell) P\bar{G}_{n,2}(X_{\ell-1}, X_\ell) - w_n(\ell, \ell) P\bar{G}_{n,2}(X_\ell, X_\ell)). \end{aligned}$$

□

5.3. Proof theorem 3.1. A2 implies that we can find $p > 1$ such that

$$\sup_{n \geq 0} \mathbb{E} \left(V_2^{2p}(X_n) \right) < \infty. \quad (29)$$

From (12), $\Gamma_{n,b}^2(h) = \sum_{\ell=1}^n \sum_{j=1}^{\ell} w_{n,b}(\ell-j)h(X_j)h(X_{\ell}) + R_n$, where $w_{n,b}(0) = n^{-1}$ and $w_{n,b}(\ell) = 2n^{-1}w_b(kc_n^{-1})$ (in particular $w_{n,b}(\ell) = 0$ for $\ell < 0$). Set $\Delta_{n,b}^{(1)}(\ell) = w_{n,b}(\ell) - w_{n,b}(\ell-1)$ and $\Delta_{n,b}^{(2)}(\ell) = 2w_{n,b}(\ell) - w_{n,b}(\ell+1) - w_{n,b}(\ell-1)$. Then Lemma 2.2 applied to $\sum_{\ell=1}^n \sum_{j=1}^{\ell} w_{n,b}(\ell-j)h(X_j)h(X_{\ell})$ gives:

$$\Gamma_{n,b}^2(h) = n^{-1} \sum_{\ell=1}^n Q_{\ell}^2 + \sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_{n,b}(\ell-j)Q_{\ell}Q_j + R_n + \zeta_n,$$

where

$$\begin{aligned} \zeta_n = & \sum_{\ell=1}^n PG(X_{\ell-1}) \sum_{j=1}^{\ell} \Delta_{n,b}^{(1)}(\ell-j)Q_j + \sum_{\ell=1}^n Q_{\ell} \sum_{j=1}^{\ell} \Delta_{n,b}^{(1)}(\ell-j+1)PG(X_{j-1}) \\ & + \sum_{\ell=3}^n PG(X_{\ell-1}) \sum_{j=1}^{\ell-2} \Delta_{n,b}^{(2)}(\ell-j)PG(X_{j-1}) \\ & + PG(X_0) \left\{ \sum_{\ell=1}^n w_{n,b}(\ell)Q_{\ell} + \sum_{\ell=1}^n \Delta_{n,b}^{(1)}(\ell)PG(X_{\ell-1}) \right\} \\ & - PG(X_n) \left\{ \sum_{j=1}^n w_{n,b}(n-j)Q_j - \sum_{j=1}^n \Delta_{n,b}^{(1)}(\ell-j)PG(X_j) \right\} \\ & + \Delta_{n,b}^{(2)}(0) \sum_{\ell=1}^n (PG(X_{\ell-1}))^2 - \left(w_{n,b}(0) - \Delta_{n,b}^{(2)}(1) \right) \sum_{\ell=1}^n PG(X_{\ell})Q_{\ell} \\ & - \Delta_{n,b}^{(2)}(1) (PG(X_n))^2 - w_{n,b}(n-1)PG(X_n)PG(X_0). \end{aligned}$$

Using A2, (29), the martingale-difference property of $\{Q_{\ell}, \ell \geq 1\}$, the smoothness of w_b , we derive that for $p > 1$ as in (29),

$$\mathbb{E}^{1/p}(|\zeta_n|^p) \leq c c_n^{-1+\frac{1}{2} \vee \frac{1}{p}}, \quad n \geq 3, \quad (30)$$

for some finite constant c .

By martingale approximation for linear partial sums (see e.g. the proof of Proposition 6.1 below), for any sequence of real numbers $\{a_{n,\ell}, 1 \leq \ell \leq n\}$,

$$\begin{aligned} \sum_{\ell=1}^n a_{n,\ell}h(X_{\ell}) &= \sum_{\ell=1}^n a_{n,\ell}Q_{\ell} + \epsilon_{n,1}, \quad \text{where} \quad \mathbb{E} \left(\left| \sum_{\ell=1}^n a_{n,\ell}Q_{\ell} \right|^{\alpha} \right) \leq c \left(\sum_{\ell=1}^n |a_{n,\ell}|^{\alpha \wedge 2} \right)^{1 \vee \frac{\alpha}{2}}, \\ \text{and } \mathbb{E}(|\epsilon_{n,1}|^{\alpha}) &\leq c \left(|a_{n,1}| + |a_{n,n}| + \sum_{\ell=2}^n |a_{n,\ell} - a_{n,\ell-1}| \right)^{\alpha}, \quad (31) \end{aligned}$$

provided $\sup_{n \geq 0} \mathbb{E}(V_2^\alpha(X_n)) < \infty$. We use (31) to bound the term R_n as given in (13) and obtain for all $n \geq 1$:

$$\mathbb{E}(|R_n|^p) \leq cn^{-p}c_n^p,$$

for some finite constant c .

By standard martingale inequalities, we obtain the bound

$$\mathbb{E} \left(\left| \sum_{\ell=1}^n \sum_{j=1}^{\ell-1} w_{n,b}(\ell-j) Q_\ell Q_j \right|^p \right) \leq c \left(\frac{c_n}{n} \right)^{\frac{p}{2}} n^{-\frac{p}{2} + 1 \vee \frac{p}{2}}.$$

□

5.4. Proof theorem 3.2. If $c_n = o(n)$ and $p \geq 2$, then from Theorem 3.1, $\Gamma_{n,b}^2(h) = n^{-1} \sum_{\ell=1}^n Q_\ell^2 + o_P(1)$. Given the ergodicity assumption $C(1, V_2^2, V_3)$ and $\sup_{k \geq 0} \mathbb{E}(V_3^q(X_k)) < \infty$, it follows from Proposition 6.1 that the term $n^{-1} \sum_{\ell=1}^n Q_\ell^2$ converges in probability to the limit

$$\int \mu(dx) \int P(x, dy) (G(y) - PG(x))^2,$$

which is easily seen to be equal to $\sigma^2(h)$. This proves the first part of the theorem.

From now on, we assume that $c_n = n$. Define $W_{n,\ell} = \frac{Q_\ell}{\sqrt{n}\sigma(h)}$. Then by (31) with $a_{n,\ell} \equiv \frac{1}{\sqrt{n}\sigma(h)}$,

$$\sum_{\ell=1}^n \frac{h(X_\ell)}{\sqrt{n}\sigma(h)} = \sum_{\ell=1}^n W_{n,\ell} + \epsilon_{n,2}, \quad \text{where } \epsilon_{n,2} \text{ converges in probability to zero.}$$

Define $\lfloor x \rfloor$ as the largest integer smaller or equal to x and for $0 \leq t \leq 1$, we introduce

$$B_n(t) = \sum_{\ell=1}^{\lfloor nt \rfloor} W_{n,\ell}, \quad \text{and } Z_n(t) = \int_0^t w_b(t-u) dB_n(u).$$

Since B_n has jumps only at times $\ell/n = \ell/c_n$, we see that $Z_n(\ell c_n^{-1}) = \sum_{j=0}^{\ell-1} w_b((\ell-j)c_n^{-1}) W_{n,j+1}$. It is also easy to see that the term R_n in (13) can be written as

$$R_n = 2B_n^2(1) \int_0^1 (1-u) w_b(u) du - 2B_n(1) \int_0^1 \left(\int_0^t w_b(u) du + \int_0^{1-t} w_b(u) du \right) dB_n(t) + \epsilon_{n,3},$$

where $\epsilon_{n,3}$ converges in probability to zero. Thus

$$\begin{aligned} \Gamma_{n,b}^2(h) &= n^{-1} \sum_{\ell=1}^n Q_\ell^2 + 2\sigma^2(h) \sum_{\ell=1}^n \frac{Q_\ell}{\sigma(h)\sqrt{n}} \sum_{j=1}^{\ell-1} w_b\left(\frac{\ell-j}{c_n}\right) \frac{Q_j}{\sigma(h)\sqrt{n}} + R_n + \zeta_n \\ &= \sigma^2(h) \sum_{\ell=1}^n W_{n,\ell}^2 + 2\sigma^2(h) \sum_{\ell=1}^n W_{n,\ell} Z_n((\ell-1)c_n^{-1}) + R_n + \zeta_n \\ &= \sigma^2(h) \sum_{\ell=1}^n W_{n,\ell}^2 + 2\sigma^2(h) \int_0^1 Z_n(t) dB_n(t) + 2B_n^2(1) \sigma^2(h) \int_0^1 w_b(u)(1-u) du \\ &\quad - 2B_n(1) \sigma^2(h) \int_0^1 g_b(u) dB_n(t) + \epsilon_{n,4}, \end{aligned}$$

where $g_b(u) = \int_0^t w_b(u)du + \int_0^{1-t} w_b(u)du$, and $\epsilon_{n,4}$ converges in probability to zero.

From the assumptions, $\sup_{\ell \geq 0} \mathbb{E}(|Q_\ell|^{2+\epsilon}) < \infty$, for some $\epsilon > 0$. Therefore, by the functional central limit theorem for martingales, $B_n \rightarrow_d B$, where $B = \{B(t), 0 \leq t \leq 1\}$ is the standard Brownian motion. By the continuous mapping theorem, $(B_n, Z_n) \rightarrow_d (B, Z)$, where $Z(t) = \int_0^t w(t-u)dB(u)$. And by the weak convergence of stochastic integrals (see, e.g., Theorem 2.2 in Kurtz and Protter (1991)),

$$\left\{ \left(B_n(t), \int_0^t Z_n(t)dB_n(t), \int_0^1 g(u)dB_n(u), \sum_{\ell=1}^n W_{n,\ell}^2 \right), 0 \leq t \leq 1 \right\}$$

converges weakly to the stochastic process

$$\left\{ \left(B(t), \int_0^t Z(u)dB(u), \int_0^1 g(u)dB(u), 1 \right), 0 \leq t \leq 1 \right\}.$$

As the remainders $(\epsilon_{n,2}, \epsilon_{n,4})$ converges in probability to 0, this entails that $\left(\sum_{\ell=1}^n \frac{h(X_\ell)}{\sqrt{n}\sigma(h)}, \Gamma_{n,b}^2(h) \right)$ converges weakly to the limit

$$\left(B(1), \sigma^2(h) \left(1 + 2 \int_0^1 Z(u)dB(u) + 2B^2(1) \int_0^1 (1-u)w(u)du - 2B(1) \int_0^1 g(u)dB(t) \right) \right).$$

The conclusion of the theorem follows by the continuous mapping theorem. $\square$

5.5. Proof Theorem 4.1. By (7), we have:

$$\sigma_n^{-1} \left[U_n(h_n) - \theta_n \binom{n}{2} \right] = \sigma_n^{-1}(n-1) \sum_{\ell=1}^n L_{n,\ell} + \sigma_n^{-1} \sum_{\ell=2}^n \sum_{j=1}^{\ell-1} Q_{n,\ell,j} + \sigma_n^{-1} \left(\zeta_n^{(1)} + \zeta_n^{(2)} \right).$$

(25) and (27) shows that $\sigma_n^{-1} \left(\zeta_n^{(1)} + \zeta_n^{(2)} \right)$ converges in probability to zero.

By conditioning on $\mathcal{F}_{\ell-1}$ and using (24) and (B1), we get

$$\sup_{1 \leq j < \ell \leq n} \mathbb{E}^{1/2} \left(|Q_{n,\ell,j}|^2 \right) \leq c \kappa_{n,2}. \quad (32)$$

Now, by the martingale property, we have

$$\begin{aligned} \mathbb{E} \left[\left(\sum_{\ell=2}^n \sum_{j=1}^{\ell-1} Q_{n,\ell,j} \right)^2 \right] &= \sum_{\ell=2}^n \mathbb{E} \left[\left(\sum_{j=1}^{\ell-1} Q_{n,\ell,j} \right)^2 \right] = \sum_{\ell=2}^n \sum_{j=1}^{\ell-1} \mathbb{E} (Q_{n,\ell,j}^2) \\ &\quad + 2 \sum_{\ell=2}^n \sum_{k=1}^{\ell-2} \sum_{j=k+1}^{\ell-1} \mathbb{E} (Q_{n,\ell,j} Q_{n,\ell,k}). \end{aligned}$$

(32) and (27) imply that $\sigma_n^{-2} \sum_{\ell=2}^n \sum_{j=1}^{\ell-1} \mathbb{E} (Q_{n,\ell,j}^2) = o(1)$, as $n \rightarrow \infty$. To deal with the second term, we fix $1 \leq k < \ell \leq n$ and define

$$T_k = T_{n,\ell,k} := \mathbb{E} \left(\sum_{j=k+1}^{\ell-1} Q_{n,\ell,j} Q_{n,\ell,k} | \mathcal{F}_{k-1} \right).$$

For $m \geq 0$, define

$$\begin{aligned}\Upsilon_{2,m}(x_{j-1}, x_{k-1}, x_k) &= \int P(x_{j-1}, dx_j) \int P^m(x_j, dx_{\ell-1}) \\ &\quad \times \int P(x_{\ell-1}, dx_\ell) \Lambda_{n,2}(x_{j-1}, x_{\ell-1}; x_j, x_\ell) \Lambda_{n,2}(x_{k-1}, x_{\ell-1}; x_k, x_\ell), \\ \Upsilon_{1,m}(x_{k-1}, x_k) &:= \int \{P^m(x_k, dx_{j-1}) - \mu(dx_{j-1})\} \Upsilon_{2,\ell-m-k-2}(x_{j-1}, x_k, x_{k-1}).\end{aligned}$$

Then almost surely we have:

$$T_k = \mathbb{E} \left(\sum_{j=0}^{\ell-k-1} \Upsilon_{1,j}(X_k, X_{k-1}) | \mathcal{F}_{k-1} \right).$$

The bound (24) and the Cauchy-Schwartz inequality imply that

$$\begin{aligned}& \left| \int P(x_{\ell-1}, dx_\ell) \Lambda_{n,2}(x_{j-1}, x_{\ell-1}; x_j, x_\ell) \Lambda_{n,2}(x_{k-1}, x_{\ell-1}; x_k, x_\ell) \right| \\ & \leq c \kappa_{n,2}^2 (\overline{W}(x_{\ell-1}, x_j) + \overline{W}(x_{\ell-1}, x_{j-1})) (\overline{W}(x_{\ell-1}, x_k) + \overline{W}(x_{\ell-1}, x_{k-1})).\end{aligned}$$

We combine this with B2 to conclude that for all $m \geq 0$,

$$|\Upsilon_{2,m}(x_{j-1}, x_{k-1}, x_k)| \leq c \kappa_{n,2}^2 \overline{\mathcal{U}}_1(x_{j-1}) \overline{\mathcal{U}}_2(x_k, x_{k-1}).$$

By the short-range dependence assumption C(1, $\overline{\mathcal{U}}_1, \overline{\mathcal{V}}_1$), it follows that for any $n \geq 0$,

$$\begin{aligned}& \left| \sum_{j=0}^{\ell-k-1} \Upsilon_{1,j}(X_k, X_{k-1}) \right| \\ & \leq \sum_{j=0}^{\ell-k-1} \left| \int \{P^j(x_k, dx_{j-1}) - \mu(dx_{j-1})\} \Upsilon_{2,\ell-j-k-2}(x_{j-1}, x_{k-1}, x_k) \right| \\ & \leq c \kappa_{n,2}^2 \overline{\mathcal{V}}_2(x_k) \overline{\mathcal{U}}_2(x_k, x_{k-1}),\end{aligned}$$

for some finite constant c . We conclude that

$$\sum_{\ell=1}^n \sum_{k=1}^{\ell-2} \sum_{j=k+1}^{\ell-1} |\mathbb{E}(Q_{n,\ell,j} Q_{n,\ell,k})| \leq c n^2 \kappa_{n,2}^2 \sup_{k \geq 1} \mathbb{E}(\overline{\mathcal{V}}_2(X_k) \overline{\mathcal{U}}_2(X_k, X_{k-1})) = o(\sigma_n^2),$$

by (27). In conclusion, $\sigma_n^{-1} \sum_{\ell=2}^n \sum_{j=1}^{\ell-1} Q_{n,\ell,j} = o_p(1)$.

Finally, we apply the martingale central limit theorem (Hall and Heyde (1980) Theorem 3.2) to conclude that the martingale array $\sigma_n^{-1} \sum_{\ell=1}^n L_{n,\ell}$ converges weakly to $\mathcal{N}(0, 1)$. Note that $L_{n,\ell} = \Lambda_{n,1}(X_\ell, X_{\ell-1})$ and also that $\sigma_{n,1}^{-2} \int \{\mu P\}(dx, dy) \Lambda_{n,1}^2(x, y) = 1$. By (23) and A??, we see that $\sup_{\ell \geq 1} \mathbb{E}^{1/2q}(|L_{n,\ell}|^{2q}) \leq c \|h_{n,1}\|_{2q, \overline{\mathcal{V}}_1}$, where $q > 1$ is as in B3. We combine this with (27) to check the Lindeberg condition: for $q > 1$ as in B3,

$$\sigma_n^{-2q} \sum_{\ell=1}^n (n-1)^{2q} \mathbb{E}(|L_{n,\ell}|^{2q}) \leq c \sigma_{n,1}^{-2q} n^{-q+1} \|\bar{h}_{n,1}\|_{2q, \overline{\mathcal{V}}_1}^{2q} \rightarrow 0,$$

by (27). To check the law of large numbers, we notice using B3 that

$$\mathbb{E}(L_{n,\ell}^2 | \mathcal{F}_{\ell-1}) = \int P(X_{\ell-1}, dz) \{\Lambda_{n,1}(z, X_{\ell-1})\}^2 \leq c \|\bar{h}_{n,1}\|_{2, \bar{V}_1}^2 \bar{W}_1^2(X_{\ell-1}).$$

Under B3, P satisfies $C(1, \bar{W}_1^2, \bar{W}_2)$ and by (27) and Proposition 6.1, we conclude that $n^{-1} \sum_{\ell=1}^n \left\{ \sigma_n^{-1} \mathbb{E}(L_{n,\ell}^2 | \mathcal{F}_{\ell-1}) \right\}$ converges in probability to one. This ends the proof of the theorem. $\square$

6. APPENDIX A: A WEAK LAW OF LARGE NUMBERS FOR MARKOV CHAINS

Proposition 6.1. *Let $\{X_n, n \geq 0\}$ be a Markov chain with invariant distribution μ and transition kernel P . Suppose that there exist measurable functions $V_1 \leq V_2 : \mathcal{X} \rightarrow [1, \infty)$ such that*

$$\sum_{k \geq 0} \|P^k(x, \cdot) - \mu\|_{V_1} \leq c V_2(x), \quad x \in \mathcal{X}, \quad (33)$$

for some finite constant c . Suppose also that $v_n := \mathbb{E}(V_2^p(X_n)) < \infty$ for each $n \geq 0$ and for some $p \in (1, 2]$. Let $\{f_n, n \geq 1\}$ be such that $f_n, P f_n \in \mathcal{L}_{V_1}$ and let $\{a_{n,k}, 0 \leq k \leq n\}$ be a sequence of real numbers such that

$$\|f_n\|_{p, V_1}^p \left(\sum_{k=1}^n |a_{n,k}| \right)^{-p} \sum_{k=1}^n |a_{n,k}|^p v_k^p \rightarrow 0,$$

$$\text{and } |P f_n|_{V_1} \left(\sum_{k=1}^n |a_{n,k}| \right)^{-1} \sum_{k=1}^n |a_{n,k} - a_{n,k-1}| v_{k-1} \rightarrow 0.$$

Then, as $n \rightarrow \infty$, $(\sum_{k=1}^n |a_{n,k}|)^{-1} \sum_{k=1}^n a_{n,k} (f_n(X_k) - \mu(f_n))$ converges in probability to zero.

Proof. Define $S_n = \sum_{k=1}^n a_{n,k} (f_n(X_k) - \mu(f_n))$ and $g_n(x) = \sum_{j \geq 0} (P^j f_n(x) - \mu(f_n))$. Under (33), $|g_n(x)| \leq c |f_n|_{V_1} V_2(x)$ and $|P g_n(x)| \leq c |P f_n|_{V_1} V_2(x)$. By the Poisson equation, $f_n(x) - \mu(f_n) = g_n(x) - P g_n(x)$ which implies that

$$\begin{aligned} S_n &= \sum_{k=1}^n a_{n,k} (g_n(X_k) - P g_n(X_{k-1})) + \sum_{k=1}^n (a_{n,k} - a_{n,k-1}) P g_n(X_{k-1}) \\ &\quad + (a_{n,0} P g_n(X_0) - a_{n,n} P g_n(X_n)). \end{aligned}$$

where the martingale array $\sum_{k=1}^n a_{n,k} (g_n(X_k) - P g_n(X_{k-1}))$ satisfies

$$\mathbb{E} \left(\left| \sum_{k=1}^n a_{n,k} (g_n(X_k) - P g_n(X_{k-1})) \right|^p \right) \leq C \|f_n\|_{p, V_1}^p \sum_{k=1}^n |a_{n,k}|^p v_k^p.$$

The last inequality follows by noting that $g_n(x) - P g_n(y) = f_n(x) - \mu(f_n) - P g_n(x) - P g_n(y)$ and by conditioning on $\mathcal{F}_{k-1}$. Thus, under the stated assumptions, $(\sum_{k=1}^n |a_{n,k}|)^{-1} S_n$ converges in probability to zero. $\square$

Remark 8. An important special case is the case where $a_{n,\ell} = 1$ and $\sup_{n \geq 0} \mathbb{E}(V_2^p(X_n)) < \infty$. In this case it is enough to have $n^{-1+1/p} \|f_n\|_{p,V_1} \rightarrow 0$. If in addition it is true that $\sup_{x \in \mathcal{X}} PV_1^p(x)/V_1^p(x) < \infty$, then clearly $\|f_n\|_{p,V_1} \leq c|f_n|_{V_1}$ and the law of large number holds if $n^{-1+1/p}|f_n|_{V_1} \rightarrow 0$.

7. APPENDIX B: A HIGHER ORDER EXTENSION

Fix $m \geq 2$ and consider a family of symmetric measurable functions $\{h_{n,m} : \mathcal{X}^m \rightarrow \mathbb{R}, n \geq 1\}$ and weights functions $\{w_{n,m} : \mathbb{N}^m \rightarrow \mathbb{R}, n \geq 1\}$. We set $\bar{\mu}_n = \bigotimes_{j=1}^m \mu$ and assume that for each n , $\bar{\mu}_n(|h_{n,m}|) < \infty$ and we define $\theta_{n,m} = \int h_{n,m}(z_{1:m}) \bar{\mu}_m(dz_{1:m})$. We consider the multivariate form

$$U_{n,m}(h_{n,m}) = \sum_{1 \leq \ell_1 \leq \dots \leq \ell_m \leq n} w_{n,m}(\ell_1, \dots, \ell_m) h_{n,m}(X_{\ell_1}, \dots, X_{\ell_m}), \quad n \geq 1.$$

We replace A1 by the following

Assumption A1' For each $r \in \{1, \dots, m\}$, there exist symmetric measurable functions $\bar{V}_r \leq \bar{W}_r : \mathcal{X}^r \rightarrow [1, \infty)$, and a finite constant c such that P satisfies $C(r, \bar{V}_r, \bar{W}_r)$ and for all $x, y \in \mathcal{X}$, $P\bar{W}(x, y) < \infty$ for all ...

For $r = 1, \dots, m$ we introduce the function

$$\bar{h}_{n,r}(x_{1:r}) := \int h_{n,m}(z_{1:m}) \left\{ \bigotimes_{j=1}^r (\delta_{x_j}(dz_j) - \mu(dz_j)) \bigotimes_{\ell=r+1}^m \mu(dz_\ell) \right\},$$

with the convention that $\bigotimes_{j=a}^b \cdot = 1$ if $a > b$. This notation is consistent with the the case $m = 2$ treated above: if $m = 2$, $\bar{h}_{n,1}(x_1) = \int h_{n,2}(x_1, z) \mu(dz) - \theta_{n,2}$, and $\bar{h}_{n,2}(x_1, x_2) = h_{n,2}(x_1, x_2) - \bar{h}_{n,1}(x_1) - \bar{h}_{n,2}(x_2) + \theta_{n,2}$, where $\theta_{n,2} = \int \int h_{n,2}(z_1, z_2) \mu(dz_1) \mu(dz_2)$. It is standard to express $h_{n,m}$ as

$$h_{n,m}(x_{1:m}) = \theta_{n,m} + \sum_{r=1}^m \sum_{1 \leq j_1 < \dots < j_r \leq m} \bar{h}_{n,r}(x_{j_1}, \dots, x_{j_r}), \quad \text{for all } x_{1:m} \in \mathcal{X}^m. \quad (34)$$

For $1 \leq j_1 \leq \dots \leq j_r \leq n$, define

$$\mathcal{S}_{m,r}(j_1, \dots, j_r) := \{(\ell_1, \dots, \ell_m) : 1 \leq \ell_1 \leq \dots \leq \ell_m \leq n,$$

$$\ell_{k_1} = j_1, \dots, \ell_{k_r} = j_r \text{ for some } 1 \leq k_1 < \dots < k_r \leq m\},$$

and define

$$w_{n,r}(j_1, \dots, j_r) := \sum_{(\ell_1, \dots, \ell_m) \in \mathcal{S}_{m,r}(j_1, \dots, j_r)} w_{n,m}(\ell_1, \dots, \ell_m).$$

Then we use (34) to write

$$U_{n,m}(h_{n,m}) = \theta_{n,m} \sum_{1 \leq \ell_1 \leq \dots \leq \ell_m \leq n} w_{n,m}(\ell_1, \dots, \ell_m) \\ + \sum_{r=1}^m \sum_{1 \leq \ell_1 \leq \dots \leq \ell_r \leq n} w_{n,r}(\ell_1, \dots, \ell_r) \bar{h}_{n,r}(X_{\ell_1}, \dots, X_{\ell_r}) = U_{n,0} + \sum_{r=1}^m U_{n,r}(\bar{h}_{n,r}), \quad (35)$$

where

$$U_{n,0} = \theta_{n,m} \sum_{1 \leq \ell_1 \leq \dots \leq \ell_m \leq n} w_{n,m}(\ell_1, \dots, \ell_m), \quad \text{and} \\ U_{n,r} = \sum_{1 \leq \ell_1 \leq \dots \leq \ell_r \leq n} w_{n,r}(\ell_1, \dots, \ell_r) \bar{h}_{n,r}(X_{\ell_1}, \dots, X_{\ell_r}).$$

The identity (35) is the Hoeffding decomposition as applied to weighted multivariate forms.

For $r = 2, \dots, m$, define

$$\bar{R}_r(x_{1:r}, dz_{1:r}) := \sum_{\ell_1 \geq 0} \dots \sum_{\ell_r \geq 0} \left\{ \bigotimes_{j=1}^r (P^{\ell_j}(x_j, dz_j) - \mu(dz_j)) \right\}, \\ \text{and } \bar{G}_{n,r}(x_{1:r}) := \int \bar{R}_r(u_{1:r}, du_{1:r}) \bar{h}_{n,r}(u_{1:r}).$$

We have

$$\int \bigotimes_{j=1}^r (\delta_{x_j}(dz_j) - P(x_j, dz_j)) \bar{G}_{n,r}(z_{1:r}) = \bar{h}_{n,r}(x_{1:r}). \quad (36)$$

(36) is the multivariate Poisson's equation for P and $\bar{h}_{n,r}$. For $r = 1$, one recovers the standard Poisson's equation and for $r = 2$, we recover (5). For $r \geq 1$ integer and a measurable symmetric function $f : \mathcal{X}^r \rightarrow \mathbb{R}$ and $1 \leq s \leq r$, we define

$$P^s f(x_1, \dots, x_r) := \int P(x_1, dz_1) \dots \int P(x_s, dz_s) f(z_1, \dots, z_s, x_{s+1}, \dots, x_r),$$

whenever the integral is well defined. Obviously, for $r = 1$, $P^1 f(x) = P f(x)$. We now introduce

$$\bar{\Lambda}_{n,r}(x_{1:r}; y_{1:r}) := \int \bigotimes_{j=1}^r (\delta_{y_j}(dz_j) - P(x_j, dz_j)) \bar{G}_{n,r}(z_{1:r}).$$

Then the Poisson's equation (36) can be written as $\bar{\Lambda}_{n,r}(x_{1:r}; x_{1:r}) = \bar{h}_{n,r}(x_{1:r})$ for all $x_{1:r} \in \mathcal{X}^r$. But observe that for $1 \leq \ell_1 \leq \dots \leq \ell_r \leq n$, if $\ell_{r-1} \leq \ell_r - 1$, then

$$\mathbb{E}(\bar{\Lambda}_{n,r}(X_{\ell_1}, \dots, X_{\ell_{r-1}}, X_{\ell_r-1}; X_{\ell_1}, \dots, X_{\ell_r}) | \mathcal{F}_{\ell_r-1}) = 0. \quad (37)$$

By the symmetry of $\bar{G}_{n,r}$ and expanding the product measure, we have

$$\bar{\Lambda}_{n,r}(x_{1:r}; y_{1:r}) = G_{n,r}(y_{1:r}) + \sum_{\alpha=1}^r \sum_{1 \leq j_1 < \dots < j_\alpha \leq r} (-1)^\alpha P^\alpha \bar{G}_{n,r}(x_{j_1}, \dots, x_{j_\alpha}, y_{-j}), \quad (38)$$

where we use the notation y_{-j} to denote the elements of $(y_1, \dots, y_r)$ with index not in $\{j_1, \dots, j_\alpha\}$. For integers $a \leq b$, and in an effort to simplify the notation, we shall use the

short-cut $X_{\ell_{a:b}}$ to denote the sequence $(X_{\ell_a}, X_{\ell_{a+1}}, \dots, X_{\ell_b})$. For $1 \leq \ell_1 \leq \dots \leq \ell_r \leq n$, and using the generalized Poisson's equation (36), we have

$$\begin{aligned} w_{n,r}(\ell_1, \dots, \ell_r) \bar{h}_{n,r}(X_{\ell_1}, \dots, X_{\ell_r}) &= w_{n,r}(\ell_1, \dots, \ell_r) \bar{\Lambda}_{n,r}(X_{\ell_{1:r}}; X_{\ell_{1:r}}) \\ &= w_{n,r}(\ell_1, \dots, \ell_r) \bar{\Lambda}_{n,r}(X_{\ell_{1:r-1}} X_{\ell_r-1}; X_{\ell_{1:r}}) + w_{n,r}(\ell_1, \dots, \ell_r) \bar{\Lambda}_{n,r}(X_{\ell_{1:r}}; X_{\ell_{1:r}}) \\ &\quad - w_{n,r}(\ell_1, \dots, \ell_r) \bar{\Lambda}_{n,r}(X_{\ell_{1:r-1}}, X_{\ell_r-1}; X_{\ell_{1:r}}). \end{aligned}$$

Using (38), to re-express the term $w_{n,r}(\ell_{1:r}) \bar{\Lambda}_{n,r}(X_{\ell_{1:r}}; X_{\ell_{1:r}}) - w_{n,r}(\ell_{1:r}) \bar{\Lambda}_{n,r}(X_{\ell_{1:r-1}}, X_{\ell_r-1}; X_{\ell_{1:r}})$, we deduce the following decomposition. Define

$$\varpi_{n,r}(\ell_1, \dots, \ell_r) := w_{n,r}(\ell_{1:r}) - w_{n,r}(\ell_{1:(r-1)}, \ell_r - 1),$$

$$\begin{aligned} T_{n,r}^{(1)}(\ell_{1:(r-1)}) &:= \sum_{\alpha=1}^r (-1)^\alpha \sum_{1 \leq j_1 < \dots < j_{\alpha-1} \leq r-1} P^\alpha \bar{G}_{n,r} \left(X_{\ell_{j_1}}, \dots, X_{\ell_{j_{\alpha-1}}}, X_{\ell_r-1}, X_{\ell_r-j} \right), \quad (39) \end{aligned}$$

$$\begin{aligned} T_{n,r}^{(2)}(\ell_{1:(r-1)}) &:= \sum_{\alpha=1}^r (-1)^\alpha \sum_{1 \leq j_1 < \dots < j_{\alpha-1} \leq r-1} P^\alpha \bar{G}_{n,r} \left(X_{\ell_{j_1}}, \dots, X_{\ell_{j_{\alpha-1}}}, X_{n-1}, X_{\ell_r-j} \right), \quad \text{and} \quad (40) \end{aligned}$$

$$T_{n,r}^{(3)}(\ell_{1:r}) := \sum_{\alpha=1}^r (-1)^\alpha \sum_{1 \leq j_1 < \dots < j_\alpha \leq r} P^\alpha \bar{G}_{n,r} \left(X_{\ell_{j_1}}, \dots, X_{\ell_{j_{\alpha-1}}}, X_{\ell_r-1}, X_{\ell_r-j} \right). \quad (41)$$

Lemma 7.1. Assume A1 and $\bar{h}_{n,r} \in \mathcal{L}_{\bar{V}_r}$ for each $n \geq 1$ and each $r \in \{1, \dots, r\}$.

$$U_{n,m}(h_{n,m}) = U_{n,0} + \sum_{r=1}^m M_{n,r} + \sum_{r=1}^m \eta_{n,r}, \quad (42)$$

where

$$M_{n,r} = \sum_{1 \leq \ell_1 \leq \dots \leq \ell_r \leq n} w_{n,r}(\ell_1, \dots, \ell_r) \Lambda_{n,r}(X_{\ell_{1:r-1}} X_{\ell_r-1}; X_{\ell_{1:r}}). \quad (43)$$

and

$$\begin{aligned} \eta_{n,r} &= \sum_{1 \leq \ell_1 \leq \dots \leq \ell_{r-1} \leq n} w_{n,r}(\ell_1, \dots, \ell_{r-1}, \ell_{r-1}) T_{n,r}^{(1)}(\ell_1, \dots, \ell_{r-1}) \\ &\quad + \sum_{1 \leq \ell_1 \leq \dots \leq \ell_{r-1} \leq n} w_{n,r}(\ell_1, \dots, \ell_{r-1}, n-1) T_{n,r}^{(2)}(\ell_1, \dots, \ell_{r-1}) \\ &\quad + \sum_{1 \leq \ell_1 \leq \dots \leq \ell_r \leq n} \varpi_{n,r}(\ell_1, \dots, \ell_r) T_{n,r}^{(3)}(\ell_1, \dots, \ell_r), \end{aligned}$$

with $T_{n,r}^{(i)}$, $i \in \{1, 2, 3\}$, as defined in (39-41).

Acknowledgments: This work is partly supported by NSF grant DMS 0906631 (Atchade) and NSF grant SES 0921505 (Cattaneo). We are grateful to Michael Jansson, Lutz Kilian and Yizao Wang for helpful discussions, comments and suggestions.

REFERENCES

- ATCHADE, Y. (2010). Kernel estimators of asymptotic variance for adaptive Markov Chain Monte Carlo. *Annals of Statistics (to appear)*.
- BHANSALI, R. J., GIRAITIS, L. and KOKOSZKA, P. S. (2007). Approximations and limit theory for quadratic forms of linear processes. *Stochastic Process. Appl.* **117** 71–95.
- DEHLING, H. and WENDLER, M. (2010). Central limit theorem and the bootstrap for U -statistics of strongly mixing data. *J. Multivariate Anal.* **101** 126–137.
- DOUC, R., FORT, G., MOULINES, E. and SOULIER, P. (2004). Practical drift conditions for subgeometric rates of convergence. *Ann. Appl. Probab.* **14** 1353–1377.
- EAGLESON, G. K. (1979). Orthogonal expansions and U -statistics. *Austral. J. Statist.* **21** 221–237.
- FLEGAL, J. M. and JONES, G. L. (2010). Batch means and spectral variance estimators in Markov Chain Monte Carlo. *Ann. Statist.* **38** 1034–1070.
- GELMAN, A., CARLIN, J. B., STERN, H. S. and RUBIN, D. B. (2004). *Bayesian data analysis*. 2nd ed. Texts in Statistical Science Series, Chapman & Hall/CRC, Boca Raton, FL.
- HALL, P. and HEYDE, C. C. (1980). *Martingale Limit theory and its application*. Academic Press, New York.
- HSING, T. and WU, W. B. (2004). On weighted U -statistics for stationary processes. *Ann. Probab.* **32** 1600–1631.
- KIEFER, N. M. and VOGELSANG, T. J. (2005). A new asymptotic theory for heteroskedasticity-autocorrelation robust tests. *Econometric Theory* **21** 1130–1164.
- KURTZ, T. and PROTTER, P. (1991). Weak limit theorems for stochastic integrals and stochastic differential equations. *Annals of Probability* **19** 1035–1070.
- MAXWELL, M. and WOODROOFE, M. (2000). Central limit theorems for additive functional of markov chains. *Annals of Probability* **28** 713–724.
- MEITZ, M. and SAIKKONEN, P. (2008). Ergodicity, mixing, and existence of moments of a class of Markov models with applications to GARCH and ACD models. *Econometric Theory* **24** 1291–1320.
- MEYN, S. P. and TWEEDIE, R. L. (2009). *Markov chains and stochastic stability*. Cambridge University Press; 2 edition, London.

- NEAVE, H. R. (1970). An improved formula for the asymptotic variance of spectrum estimates. *Ann. Math. Statist.* **41** 70–77.
- PRIESTLEY, M. B. (1981). *Spectral analysis and time series. Vol. 1.* Academic Press Inc. [Harcourt Brace Jovanovich Publishers], London. Univariate series, Probability and Mathematical Statistics.
- ROBERTS, G. O. and ROSENTHAL, J. S. (2004). General state space Markov chains and MCMC algorithms. *Probab. Surv.* **1** 20–71 (electronic).
URL <http://dx.doi.org.proxy.lib.umich.edu/10.1214/1549578041000000024>
- SERFLING, R. J. (1980). *Approximation Theorems of Mathematical Statistics.* Wiley, New York.
- WU, W. B. and SHAO, X. (2007). A limit theorem for quadratic forms and its applications. *Econometric Theory* **23** 930–951.
- YOSHIHARA, K.-I. (1976). Limiting behavior of U -statistics for stationary, absolutely regular processes. *Z. Wahrscheinlichkeitstheorie und Verw. Gebiete* **35** 237–252.